ElderCareAI

An AI-Driven IoT System for Elderly Monitoring and Caregiver Support

1. Implementation Budget and Cost Specification

The following budget covers one development laptop, one Android phone, one Galaxy Watch6, one month of internet and electricity, and a project food allowance. Amounts are expressed in Philippine pesos (PHP).

Table 1. Estimated implementation cost

Item	Cost basis	Estimated amount
Laptop	One unit	₱30,000.00
Android phone	One unit	₱14,000.00
Samsung Galaxy Watch6	One unit	₱4,500.00
Hardware subtotal		**₱48,500.00**
Visual Studio Code and Android Studio	No-cost development tools	₱0.00
React Native, Expo and required SDKs	Prototype development	₱0.00
Supabase	Free-plan allowance	₱0.00
Firebase FCM and Expo Push Service	No-cost messaging services	₱0.00
Gemini API and Expo build services	Within available no-cost quotas	₱0.00
Software subtotal		**₱0.00**
Internet connection	One month	₱1,699.00
Electricity	One month	₱3,000.00
Food allowance	Project allowance	₱2,000.00
Operating and miscellaneous subtotal		**₱6,699.00**
TOTAL IMPLEMENTATION BUDGET	**One-month budget basis**	**₱55,199.00**

The laptop and phone amounts are retained from the proposed budget. The watch, internet and electricity amounts reflect the revised project estimates. The food allowance is treated as a single project expense.

Internet and electricity total ₱4,699.00 per month. For a longer implementation period, add ₱4,699.00 for each additional month. The one-month total does not include labor, repairs, store-publishing fees, or paid cloud/API usage beyond the budgeted no-cost allowances.

Zero-cost software entries describe the prototype budget, not unlimited service capacity or an open-source license for every product. Review provider quotas before expanding the system.

Technical references: Supabase pricing; Firebase pricing; Expo service costs; Expo Push Service cost; Gemini API pricing. Accessed 3 November 2026.

2. Software Specification

ElderCareAI uses a custom Android mobile application, a native Wear OS watch application, a web administration portal, and a Supabase cloud backend. Galaxy Watch6 supplies supported sensor and GPS updates. The phone imports sleep and recorded oxygen history through the Samsung Health Data SDK.

Table 2. Software development and deployment requirements

Component	Technology	Purpose
Mobile application	React Native, Expo and TypeScript / JavaScript	Caregiver dashboard, live readings, map, care plans, alerts, reports and Elle AI.
Mobile build	Custom Android APK with native modules	Includes Samsung data access and Android push support. Expo Go cannot load these custom native integrations.
Watch application	Kotlin, Android SDK and Wear OS	Collects supported sensor and GPS updates; provides monitoring controls, spot oxygen measurement, SOS and medication actions.
Health integrations	Samsung Health Sensor SDK and Samsung Health Data SDK	The watch SDK supplies supported sensor readings. The phone SDK imports available sleep and recorded oxygen history.
Database and cloud services	Supabase PostgreSQL, Auth, Realtime and Storage	Stores accounts, patient records, vital events, location updates, care schedules and alerts; synchronizes authorized views.
Server-side processing	TypeScript in Supabase Edge Functions; PostgreSQL SQL functions	Uses a Deno-compatible Edge Runtime, with database rules, scheduled work and notification queues.
Notifications	Expo Notifications, Expo Push Service and Firebase Cloud Messaging v1	Delivers Android push alerts and care reminders; uses Android sound channels and local reminders where configured.
AI assistance	Gemini API through server-side functions	Elle answers caregiver questions using saved patient context. Historical fact queries use stored records; statistical trends support summaries.
Administration portal	HTML, CSS and JavaScript in a modern browser	Authorized administration of accounts, patient records, stored monitoring data, reports and audit history.
Development tools	Visual Studio Code, Android Studio, Android SDK, Gradle, JDK and Node.js	Supports mobile/web development and APK builds. Node.js is development tooling; the Edge Functions runtime is Deno-compatible.

Deployment requirements. Install the phone and watch applications with compatible signing credentials. Configure Google sign-in, Samsung SDK access, backend secrets and Firebase/Expo credentials. Developer testing and public distribution have different Samsung SDK access requirements; production use must satisfy Samsung's applicable registration process.

Technical references: Expo custom native code; Supabase Edge Functions; Expo FCM v1 credentials. Accessed 3 November 2026.

3. Hardware Specification

The development computer, caregiver phone and patient watch perform different functions and therefore have separate requirements. The following development baseline is intended for a physical-device testing workflow; emulator use requires additional resources.

Table 3. Development computer requirements

Component	Minimum baseline	Recommended configuration
Operating system	64-bit Windows 10 supported by the development tools	A currently supported 64-bit Windows release
Processor	Compatible 64-bit processor; virtualization support for emulator use	Recent Intel Core i5 / AMD Ryzen 5 class or better
Memory	8 GB RAM for Android Studio without an emulator	16 GB for physical-device work; 32 GB for heavier emulator use
Storage	Space for Android Studio, SDKs, Gradle caches and project files	SSD with at least 32 GB free for tools, plus project and build space
Display and connectivity	1280 × 800 display; internet and USB or wireless debugging	1920 × 1080 or higher; stable internet and USB access

Table 4. Runtime devices and infrastructure

Device / service	Requirement
Caregiver Android phone	Android 10 / API 29 or later; Bluetooth, Wi-Fi or mobile data, location services and notification support. Google Play services are needed for FCM delivery. Android 13 or later is preferred for current permission controls.
Patient smartwatch	Samsung Galaxy Watch6 with a supported Wear OS / Health Sensor Service version, the ElderCareAI watch APK, enabled sensor and location permissions, and adequate charge.
Health history on phone	Samsung Health 6.30.2 or later and Galaxy Wearable, with the patient's watch paired to the same phone. Grant the required sleep and blood-oxygen read access.
Administrator workstation	A laptop or desktop with a modern web browser and a secure internet connection. A dedicated on-premises database server is not required.
Network and power	Internet access for uploads and cloud services; Bluetooth for watch–phone features where used. Provide the proper phone/watch chargers. Active sensors and GPS increase battery use.

Sensor scope. The watch provides optical heart-rate/oxygen sensing, wrist skin-temperature sensing, motion sensing and GPS. Sleep and stress are derived records or calculations rather than separate physical sensors. GPS accuracy depends on permissions, signal conditions and a valid location fix. Compatibility with other watch models requires separate testing.

Technical references: Android Studio system requirements; Samsung Health Data SDK requirements. Accessed 3 November 2026.

4. Human Resource Specification

ElderCareAI supports three user groups. Access to patient information depends on the authenticated user's role and assigned records.

4.1 Caregiver

The caregiver manages the assigned older adult's profile, pairs the watch, reviews readings and location, maintains medications and appointments, responds to alerts, and consults Elle AI. The caregiver also enables notification permissions and checks whether displayed data is current.

4.2 System Administrator

The administrator uses the web portal to manage authorized user accounts and patient records, review stored monitoring data and audit activity, and generate system reports. Administrative privileges must be restricted to approved personnel.

4.3 Older Adult / Watch Wearer

The older adult wears the Galaxy Watch6, keeps it charged, and uses the watch's monitoring, spot oxygen, SOS and medication controls as needed. The caregiver can assist with setup and permissions. A separate caregiver-style phone account is not required simply to wear the paired watch.

5. Program Specification

5.1 Functional Requirements

FR-01. Authentication and profiles. Provide caregiver Google sign-in and separate authenticated administrator access. Maintain caregiver, patient and emergency-contact details with role-based access.

FR-02. Watch integration. Pair the Galaxy Watch6 with the correct patient, request the necessary permissions, display connection status and provide monitoring start/stop controls.

FR-03. Health monitoring. Display supported heart rate, oxygen, wrist skin temperature, steps, sleep and derived stress/HRV information with units and individual measurement times.

FR-04. Vital-event history. Save every accepted new vital event with its original time and source. Retain unacknowledged watch samples for retry and prevent duplicate retry records. Missing metrics remain blank rather than being fabricated.

FR-05. Location tracking. Display the patient's latest valid watch location and update the map when new positions arrive. Indicate location age or accuracy so an old fix is not presented as current movement.

FR-06. Medications and appointments. Create, update and display care schedules, instructions and reminders. Synchronize supported medication "Taken" actions between watch and app, with clear save and reminder-setup feedback.

5.1 Functional Requirements (continued)

FR-07. Threshold alerts. Evaluate valid uploaded readings against the configured app review rules. Save threshold warnings in Alerts and Notifications, apply duplicate/cooldown controls, and queue caregiver push delivery.

FR-08. SOS and notifications. Allow a watch SOS request to create a caregiver alert. Deliver SOS, vital warnings, medication reminders and appointment reminders through the configured notification paths. SOS is an assistance request, not an automatic emergency-services dispatch.

FR-09. Health history import. Import permitted sleep and recorded oxygen data from the selected patient watch through the phone's health history integration. Retry failed uploads without inventing or duplicating readings.

FR-10. Elle AI and analytics. Answer questions about saved patient history, summarize trends and provide clearly qualified statistical insights. Distinguish missing data from a confirmed negative result; avoid diagnosis or guaranteed health predictions.

FR-11. Reports. Provide patient vital-history and medication-schedule PDF exports, together with authorized administrative reporting.

FR-12. Administration and settings. Support account and record management, audit review, device status, notification preferences and sound settings. Expose clear permission, connectivity and synchronization errors.

5.2 Data Collection and Update Behavior

Table 5. Availability of monitored data

Data	Collection and update behavior
Heart rate	Updates from valid delivered watch sensor events while monitoring. New accepted readings are logged individually, including values above the warning threshold.
Blood oxygen (SpO_2)	Manual spot measurements and optional automatic spot checks while the watch app is in the foreground. Recorded overnight oxygen is imported when available; this is not a continuous background oxygen stream.
Wrist skin temperature	Uses valid supported skin-contact readings. This value is wrist skin temperature, not core body temperature.
HRV / stress level	Rolling RMSSD HRV is calculated from valid beat intervals over about 30–60 seconds. A personal reference supports an experimental 0–100 stress estimate; no score is invented while the baseline is insufficient.
Steps	Uses step-counter updates while monitoring. Time when monitoring is off cannot be assumed to have complete step coverage.
Sleep	Imports actual sleep sessions after they are available through the health-history SDK. It does not stream second by second or guarantee delivery immediately on waking.

Live updates are limited by sensor delivery, signal quality, connection state and upload success. Each vital retains its own measurement time. The stress estimate is experimental and not medical grade.

Technical references: Samsung sensor data and on-demand limitations. Accessed 3 November 2026.

5.3 Automatic Vital Review Rules

The following are the current application review rules. They are not individualized medical limits and should not be presented as a diagnosis or as proof that a patient is safe.

Table 6. Configured threshold warnings

Metric	Warning condition	Evaluation basis
Heart rate	Below 50 BPM or above 110 BPM	A valid new recorded sample. Exactly 110 BPM does not meet the high-rate rule.
SpO_2	Below 95%	An accepted spot or imported oxygen reading; continuous coverage is not implied.
Sleep duration	Recorded session below 5 hours	Evaluated after the sleep record becomes available and is imported.
Wrist skin temperature	Change of at least 0.8°C AND at least 2 standard deviations	Requires at least five recent same-watch reference readings; uses up to seven readings within 24 hours.
HRV, stress and steps	No automatic threshold configured	Displayed and recorded where available; these values do not currently generate automatic threshold warnings.

A qualifying reading can create a warning after upload. Repeated notices for an active live rule are limited to one per hour; a return to range and a new crossing reset it. Imported records use deduplication. Push work is triggered after database commit, with scheduled recovery. Android and network conditions affect delivery.

5.4 Nonfunctional Requirements

NFR-01. Data quality. Validate values, timestamps and sensor quality. Mark missing or stale data clearly. Consumer wearable readings and derived scores must not be represented as clinical-grade measurements.

NFR-02. Security and privacy. Use authenticated access, role checks, database row-level policies and encrypted network communication. Keep privileged backend, AI and push credentials out of client code.

NFR-03. Reliability and integrity. Use event identifiers, acknowledgments and retries to reduce data loss and duplicate history. Preserve independent measurement times and source details.

NFR-04. Responsiveness. Display new accepted data and queue threshold alerts promptly. Sensor intervals, continuous coverage and notification arrival times cannot be guaranteed.

NFR-05. Background operation. Support Android push when the app is not open, using the correct APK, credentials, registration and permission. Force-stop, offline state, battery policies, muted channels and Do Not Disturb can affect delivery or sound.

NFR-06. Usability and compatibility. Use readable labels, units, clear controls and understandable errors. Validate supported Android phones, Galaxy Watch6 and modern-browser administrator access.

NFR-07. Maintainability and capacity. Keep mobile, watch and backend modules organized and testable. Monitor database growth, cloud quotas, failures and retry queues as the number of patients and recorded samples increases.

6. Proposed Module Assignments

This proposed allocation covers the updated Galaxy Watch6 system, health-history import, Elle AI and notifications. Each proponent owns six modules for implementation and testing, for a total of 18 modules.

Table 7. Module ownership and user participation

Proponent	Assigned module	Watch wearer	Caregiver	Admin
Kristoffer James G. Layos	Landing page and role dashboards	—	Yes	Yes
	Login, authentication and role control	—	Yes	Yes
	Profiles and settings	—	Yes	Yes
	Caregiver account management	—	—	Yes
	Administrator access management	—	—	Yes
	Security and audit logs	—	—	Yes
John Wayne C. Planos	Watch pairing and sensor integration	Yes	Yes	—
	Live vital signs and activity	Yes	Yes	—
	Oxygen measurement and sleep history	Yes	Yes	—
	HRV and experimental stress estimate	Yes	Yes	—
	Medication and appointment care	Yes	Yes	—
	Health history and PDF reports	—	Yes	Yes
Melvence R. Bagonggong	SOS and caregiver response	Yes	Yes	—
	Automatic vital-threshold alerts	—	Yes	Yes
	GPS tracking and map updates	Yes	Yes	—
	Caregiver push and reminder notifications	—	Yes	Yes
	Elle AI and health analytics	—	Yes	—
	System reports and device-status monitoring	—	—	Yes

“Yes” indicates use or participation, including wearing the watch. It does not grant permissions. A dash indicates no primary participation. Role totals overlap because modules can serve several groups.

Table 8. Module participation totals (one point per included module)

Proponent	Modules	Watch wearer	Caregiver	Admin
Kristoffer James G. Layos	6	0	3	6
John Wayne C. Planos	6	5	6	1
Melvence R. Bagonggong	6	2	5	3
TOTAL	**18**	**7**	**14**	**10**